

Multiple Choice

1. **Answer:** D

Options: A) High metabolic rate to generate heat; B) Storage of water in fatty tissues; C) Long loops of Henle to maximize water reabsorption; D) Short loops of Henle to maximize water secretion; E) Large amounts of body hair to trap heat

Note: Correct option: D - Short loops of Henle to maximize water secretion

2. **Answer:** A

Options: A) Courtship is primarily for entertainment purposes among terns; B) Courtship is not necessary for reproduction; C) Courtship is a way to show dominance; D) Courtship is a random behavior with no impact on reproduction; E) Courtship is necessary for physical transformation required for reproduction

Note: Correct option: A - Courtship is primarily for entertainment purposes among terns

3. **Answer:** A

Options: A) Oxygen is used to maintain the temperature of cells.; B) Oxygen is used by cells to produce alcohol.; C) Oxygen is used to produce carbon dioxide in cells.; D) Oxygen is solely responsible for stimulating brain activity.; E) Oxygen is used to synthesize proteins in cells.

Note: Correct option: A - Oxygen is used to maintain the temperature of cells.

4. **Answer:** D

Options: A) O₂; B) CH₄ (Methane); C) CO₂; D) H₂O; E) N₂ (Nitrogen)

Note: Correct option: D - H₂O

5. **Answer:** A

Options: A) Duplication of chromosome 14; B) Translocation between chromosomes 14 and 21; C) Deletion in chromosome 21; D) Mutation in chromosome 14; E) Duplication of chromosome 18

Note: Correct option: A - Duplication of chromosome 14

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'pale, small, short, and weak'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 5% chance. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Coevolution. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key